

IT517 Pattern Recognition, Spring 2025

Term Project

- (a) Write a function to generate random samples from a univariate normal distribution $N(\mu, \Sigma)$.
(b) Write a function to calculate the discriminant function of the following form
$$g_i(x) = -\frac{1}{2}(x - \mu)^t \Sigma_i^{-1}(x - \mu) - \frac{d}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma_i| + \ln P(w_i)$$
For a given normal distribution and prior probability $P(w_i)$
(c) Write function to calculate the Euclidian distance in any dimension between two arbitrary points.
(d) Write a function to compute the covariance matrix of a given dataset in any dimension.
(e) Write function to calculate the Mahalanobis distance between the mean μ and a arbitrary point x given the covariance matrix Σ . Compare the output of your function with Matlab or other standard library functions.
- Implement the Perceptron algorithm and demonstrate the performance of your implementation using some datasets.
- Implement the backpropagation algorithm and demonstrate the performance of your implementation using some datasets.
- Implement the c-means clustering algorithm. Validate the results of your implementation on some suitable dataset.
- Implement the agglomerative hierarchical clustering algorithm. Validate the results of your implementation on some suitable dataset.
- Implement SFS and GSFS algorithm for feature selection and compare their performance on some datasets.
- Implement SBS and GSBS algorithm for feature selection and compare their performance on some datasets.
- Implement the KNN classifier and discuss the results of your implementation on some datasets using different values of the parameter k.
- Implement PCA. Do a clustering on some datasets using an available clustering tool/function using the original attributes and PCA attributes and discuss the results.

Sl. No. 1 is compulsory for all one of the remaining as per the following table:

Roll No.	CSM24001	CSM24004	CSM24008	CSM24036	CSM24042	CSM24051	CSM24055	CSM24064	CSM25003
Sl. No	1	2	3	4	5	6	7	8	9